Domain and Range Name ______________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $[-4, \infty)$ # ______ State the domain of $f(x) = \dfrac{x+1}{2-x}$ in interval notation.	ANSWER: $[-3, 3) \cup (3, \infty)$ # ______ State the domain of $f(x) = \sqrt{25 - x^2}$ in interval notation.
ANSWER: $(-\infty, \infty)$ # ______ State the domain of $f(x) = \dfrac{1}{x^2 - 9}$ in interval notation.	ANSWER: $(-\infty, 2) \cup (2, \infty)$ # ______ State the domain of $g(x) = \sqrt{7 - x}$ in interval notation.

ANSWER: $(-\infty,-3)\cup(-3,3)\cup(3,\infty)$

#________ Write the inequality $x \leq -2$ in interval notation.

ANSWER: $(-\infty,7]$

#________ State the domain of $h(x) = \dfrac{5}{\sqrt{x-3}}$ in interval notation.

ANSWER: $[-5,5]$

#________ State the range of $f(x) = \sqrt{25-x^2}$ in interval notation.

ANSWER: $(-\infty,-2]$

#________ State the domain of $f(x) = \sqrt{x+4}$ in interval notation.

ANSWER: $[0,5]$

#________ State the domain of $f(x) = x^2-3x+2$ in interval notation.

ANSWER: $(3,\infty)$

#________ State the domain of $f(x) = \dfrac{\sqrt{x+3}}{x-3}$ in interval notation.